

MTH 204: Symmetry

Instructor: Dr Tanusree Khandai

February 4, 2026

Midsem 1

Time: 1 hour

Total marks: 20

- ✓(i) (4 marks) Let A be a 2×2 real matrix and let

$$f_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

be defined by

$$f_A \left(\begin{pmatrix} a \\ b \end{pmatrix} \right) = A \left(\begin{pmatrix} a \\ b \end{pmatrix} \right).$$

Which of the following matrices define an isometry of $\mathbb{R}^2$?

$$\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ -2 & -1 \end{pmatrix}.$$

Justify your answer.

- ✓(ii) (4 marks) Let

$$SL_n(\mathbb{R}) = \{A \in GL_n(\mathbb{R}) : \det A = 1\}.$$

Show that $SL_n(\mathbb{R})$ is a subgroup of $GL_n(\mathbb{R})$.

- (iii) (4 marks) Let

$$H = \{x \in \mathbb{R}^\times : x^2 \in \mathbb{Q}\}.$$

Prove that H is a subgroup of the multiplicative group $\mathbb{R}^\times$.

- ✓(iv) (4 marks) State the number of isometries (symmetries) of a regular tetrahedron. Justify your answer. Determine the inverse of the symmetry (214)(341)(42).

- ✓(v) (4 marks) The points $(2, 0)$, $(0, 2)$, $(-2, 0)$, and $(0, -2)$ determine a square $ABCD$ in $\mathbb{R}^2$.

- Find the matrices corresponding to reflections across the diagonals (AC) and (BD) .
- Verify that these matrices are orthogonal.

ul.
= 2x3x4
= 24